Wyatt Brown

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Summary

Data Science undergraduate at the University of Minnesota with a strong foundation in statistical theory, machine learning, and optimization. Actively seeking internships and research opportunities to apply quantitative methods to real-world problems.

Education

University of Minnesota | Minneapolis, MN
Bachelor of Science in Data Science (3.5 GPA) | Expected December 2026

- **Relevant Coursework:** Optimization for Machine Learning (Python, Julia/JuMP), Operations Research for Data Science (Case Studies), Regression & Statistical Computing, Theory of Statistics I, Machine Learning Fundamentals, Applied Multivariate Methods, AI Under Uncertainty (Bayesian Networks, RL), Software Development Processes
- **Upcoming Coursework:** Statistical Machine Learning, Theory of Statistics II, GenAI for Software Engineering, Systems Engineering I

Skills

- **Languages:** Python, R, Java
- **Libraries & Frameworks:** JAX (Flax, Optax), PyTorch, Scikit-learn, Pandas, NumPy
- **Tools & Platforms:** VSCode, Docker, Git, Postman, Jupyter Notebooks, Excel, Adobe Suite
- **Assured Public Speaker:** Articulating complex ideas to wide audiences

Core Competencies

- **Quantitative & Statistical Foundations:** Rigorous academic training in Multivariable Calculus, Linear Algebra, and statistical theory, leveraging mathematical principles to engineer machine learning models, perform multivariate analysis, and solve complex optimization problems.
- **Machine Learning & Probabilistic AI:** Proficient in developing machine learning models and implementing probabilistic frameworks, including Bayesian networks and reinforcement learning algorithms.
- **Software Development Lifecycle:** Experienced in managing the full software lifecycle, from initial requirements to deployment, utilizing Git-based version control for streamlined, collaborative development workflows.

Data Science & Machine Learning Projects

Dockerized Particle Simulator with Python API

- Developed an interactive particle simulator in Java with a CSV backend to manage object-state persistence, facilitated by a Python Flask API enabling dynamic updates and state-saving.
- Containerized the application using Docker in two configurations, successfully isolating the API and the full application stack to ensure portability and streamlined deployment.

Vending Machine Sales Optimization Model

- Engineered a "Predict-then-Optimize" framework using Python (CART) and Julia (JuMP) to automate stocking decisions for a fleet of 239 university machines.
- Developed a "Sales Velocity" feature to correct temporal bias and implemented an Integer Linear Program (ILP) to minimize economic regret.
- Achieved a 51% reduction in expected economic costs per machine by optimizing product assortment based on location and seasonality.

Personal Portfolio Website | wrbr0wn.github.io / wyattbrown.info

- Designed, developed, and deployed a portfolio website using HTML and CSS as a central hub for professional links and project showcases.
- Developed an integrated digital art gallery sub-page featuring a Lightbox modal viewer for high-resolution image display, demonstrating end-to-end feature development within an existing codebase.

Professional Experience

XLR8 Sports Training LLC | Menasha, WI
Freelance IT Consultant & Business Operations | Oct 2022 – Jan 2025

- Provided full-stack IT support for computer systems, networks, printers, and a self-installed security system.
- Spearheaded the complete migration of 5+ years of business records from analog files to secure digital storage.
- Managed client-facing operations for 50+ clients, including scheduling, payment processing with QuickBooks, and onboarding.